

Geometric Distribution

The geometric distribution describes the probability that the first success in a sequence of independent Bernoulli trials occurs after a certain number of failures.

Probability Mass Function (PMF)

Let X be the number of failures before the first success. The probability that there are exactly x failures before the first success is:

$$P(X = x) = q^x p, \quad x = 0, 1, 2, \dots$$

Here, p is the probability of success and $q = 1 - p$.

Mean and Variance

$$E[X] = \frac{q}{p}$$
$$\text{Var}[X] = \frac{q}{p^2}$$

Variance $\times p$ = mean. So, variance $\geq$ mean.

Memoryless Property

The geometric distribution is memoryless, i.e., the probability of future failures does not depend on how many failures have already occurred.

Mathematically, for any non-negative integers m and n :

$$P(X > m + n \mid X > m) = P(X > n)$$

This implies that even after m consecutive failures, the probability of waiting at least n more trials for the first success remains the same.

Exercise

1. A coin is tossed repeatedly until a head appears. Find the probability that the first head appears on the fifth toss.
2. A basketball player has a free-throw success probability of $p = 0.8$. Find the probability that she misses exactly four shots before making her first successful throw.
3. A student answers multiple-choice questions randomly, each with a 25% chance of being correct. What is the probability that the student gets the first correct answer on the sixth question?
4. A quality inspector finds that the probability a product is defective is $p = 0.1$. Let X denote the number of non-defective products checked before finding the first defective one. Find: The expected number of non-defective items before the first defective one. Also, find the variance of X .
5. In a transmission system, the probability of successfully sending a packet is $p = 0.7$. Find: (i) the mean number of failed transmissions before the first success, (ii) the standard deviation.
6. A person is playing a dice game where rolling a 6 means winning. Let X denote the number of failures (non-6 outcomes) before the first 6 appears. Compute $E[X]$ and $\text{Var}(X)$.
7. A machine has a 20% chance of failing in each test run. Given that the machine has already passed 4 tests without failure, what is the probability that it will pass at least three more tests before failing?
8. Suppose $X \sim \text{Geometric}(p = 0.4)$. Find $P(X > 7 \mid X > 3)$ using the memoryless property.
9. A web server crashes with probability $p = 0.02$ on any given request. Let X be the number of successful requests before the first crash. (i) Find $E[X]$ and $\text{Var}(X)$. (ii) What is the probability that the server can handle at least 100 requests before crashing?
10. In a production line, the probability that a defective item appears is $p = 0.05$. A supervisor inspects items until the first defective one is found. If it is known that the first defective did not appear in the first 10 items, what is the probability that it does not appear in the next 5 items?